UC BERKELEY EECS · AGENT EVALUATION & INFRASTRUCTURE

Tasks and infrastructure for agent post-training.



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IN ONE SLIDE

Two things I do well.

1 · Build tasks that train models

ALE co-first author. **Lead** of the SWE, Medical, and Bio tracks; **co-lead** of Game, Physics, and Astronomy — from expert sourcing to programmatic graders and verifiable rewards.

2 · Build the infra agents run on

Built Equile: sandboxed multi-harness agent runtime — isolation, snapshot pause/resume, scheduling — the machinery an RL rollout fleet needs.

AGENTS' LAST EXAM · ARXIV:2606.05405 · UNDER REVIEW, NEURIPS 2026

Measuring frontier agents on work that matters.

- **1K+ expert-designed, long-horizon tasks** across 55 subdomains — real economic work, not puzzles.
- Every task is an **executable environment**: `load / start / evaluate`, sandboxed VM, computer-use included.
- **Programmatic graders** — no hand labels in the scoring loop; scores double as **verifiable rewards (RLVR)**.

1,000+	55	#1	310
executable tasks	subdomains	HF paper of the week	authors — I'm co-first

My role: led the task pipeline and auto-scoring; run evals with frontier model teams.

ALE · MY TRACKS

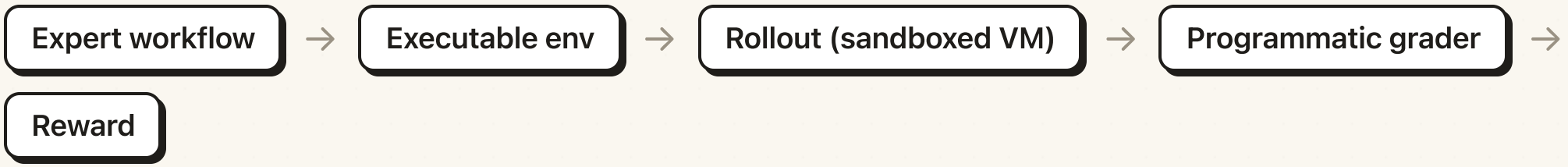
I own task construction, domain by domain.

Lead SWE Medical Bio

Own the track end to end: recruit domain experts, turn their workflows into executable task specs, build the graders, calibrate difficulty, QC every task that ships.

Co-lead Game Physics Astronomy

Same pipeline, shared ownership. Related output: **Agent's GameDev Exam** (ICLR 2027, under review) and **ScienceIDE** (HF #2 of the day).



TASK CONSTRUCTION · WHAT I OPTIMIZE FOR

A task is only as good as its reward.

- **Verifiable, not judged**: programmatic graders (hash · tabular · geometric · visual · behavioral) — no hand labels, no LLM-judge drift in the reward signal.
- **Hard to hack**: graders check the artifact and the trace; checks keep them trustworthy as models get stronger.
- **Long-horizon by design**: real expert workflows (hours of work), computer-use included — the regime where post-training has headroom.
- **Trajectory analysis closes the loop**: we run frontier models (OpenAI, Anthropic, Meta TBD, Qwen, Seed) and report failure modes back — I know where tasks break models and where models break tasks.

EQUILE · INDEPENDENT PROJECT · APR–OCT 2026

Agent infra, built from zero.

- **One session API, many harnesses**: Claude Code, Codex, Cursor, custom Docker — same contract for workspace FS, tool/MCP I/O.
- **Isolation that scales**: per-session sandboxes with multi-tenant packing (>20× per-agent compute reduction); auth never in images or CLI args.
- **Snapshot pause/resume** and background scheduling: atomic claims, lease reclaim — agents scheduled like model inference.
- **Why it matters here**: an RL rollout fleet is the same problem — thousands of isolated environments, reproducible resets, live monitoring, nothing leaking between runs.

THE FIT

What I'd do for post-training.

1. Scale task production: I have run the full pipeline — experts → executable envs → graders → verifiable rewards — across six domains; I can stand it up for yours.
2. Keep rewards honest: grader QC, anti-hacking checks, trajectory-level failure analysis — the difference between a number and a training signal.
3. Build the rollout infra: sandboxed, reproducible, schedulable environments at fleet scale — I have built this once already, solo.

THANK YOU

Let's talk.

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ALE: agents-last-exam.org · arXiv:2606.05405 / ScienceIDE: arXiv:2609.19134